



TalentBattle

LOGICAL REASONING

2019



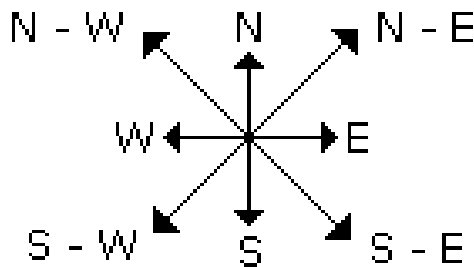
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1.DIRECTION SENSE TEST

- **There are four main directions:** North (N), East (E), West (W), South(S)
- **There are Four Cardinal Directions as shown in the diagram:**

North-East (NE), North-West (NW), South-East (SE), South-West (SW)



- Angle between North and East is 90° (Clockwise)
- Angle between North and Northeast is 45° (Clockwise)

- **Shadow's position:**

Morning (or) Sunrise (or) Dawn – WEST

Evening (or) Sunset (or) Dusk – EAST

Noon – No Shadow

Some Important Points:

1. At the time of sunrise if a man stands facing the east, his shadow will be towards west.
2. At the time of sunset the shadow of an object is always in the east.
3. If a man stands facing the North, at the time of sunrise his shadow will be towards his left and at the time of sunset it will be towards his right.
4. At 12:00 noon, the rays of the sun are vertically downward hence there will be no shadow.

Types of Problems:

- Facing Directions
- Distance Travelled
- Shadow Based Problems

1) Ramesh is facing North. He turns 135° in anti clock direction and then 180° in clock wise direction. Which direction is he facing now?

- a) South East b) South c) North East d) North

Answer: c) North East

Explanation:

135° (anticlockwise) - 180° (clockwise) = 45° (clockwise). He turns 45° clockwise from North.

2) Neha starts walking towards west. After walking 125 meter, she turns to the left and walks 65 meter straight. Again she turns to the left, walks a distance of 85 meters straight, again she turns to the left and walks a distance of 65 meters. How far is she from starting point?

- a) 45 meter b) 75 meter c) 40 meter d) 180 meter

Answer: c) 40 meter

Explanation:

125 - 85 = 40 meter from starting point

3) A man turns 4km towards west, then he turns right and travels 3km. What is the shortest distance between his start point and end point?

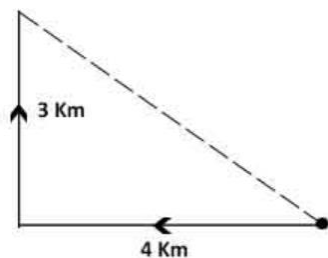
Solution: By Pythagoras theorem,

$(\text{Shortest distance})^2 = 4^2 + 3^2 = 16 + 9 = 25 \Rightarrow 5\text{km}$

4) One morning after sunrise Amit while going to college met Narendra at road crossing. Narendra's shadow was exactly to the right of Amit. If they were face to face, which direction was Amit facing? a) South b) North c) East d) West

Answer: a) South

Explanation:

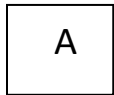


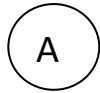
In morning the shadow falls towards the west. Narendra's shadow falls to the right of the Amit. So Amit is facing South.

2. BLOOD RELATIONS

Mother's or Father's Father	Grand father
Mother's or Father's Mother	Grand mother
Mother's or Father's Son	Brother
Mother's or Father's Daughter	Sister
Mother's or Father's Brother	Uncle
Mother's or Father's Sister	Aunt
Husband or Wife's sister & Brother's wife	Sister in law
Husband or Wife's brother & Sister's husband	Brother in law
Son's Wife	Daughter in law
Daughter's Husband	Son in law
Brother's or Sister's Son	Nephew
Brother's or Sister's Daughter	Niece
Uncle or Aunt's Son/Daughter	Cousin
Grandson's or Granddaughter's Daughter/Son	Great Grand Daughter/Son
Husband or Wife	Spouse
"Real" brother or sister	Siblings

There is a standard set of symbols which help one decode family relationship, Below is list of such symbols. Family Tree Notations:

 ---> A is Male

 ---> A is Female

A ----- B ---> A and B are siblings

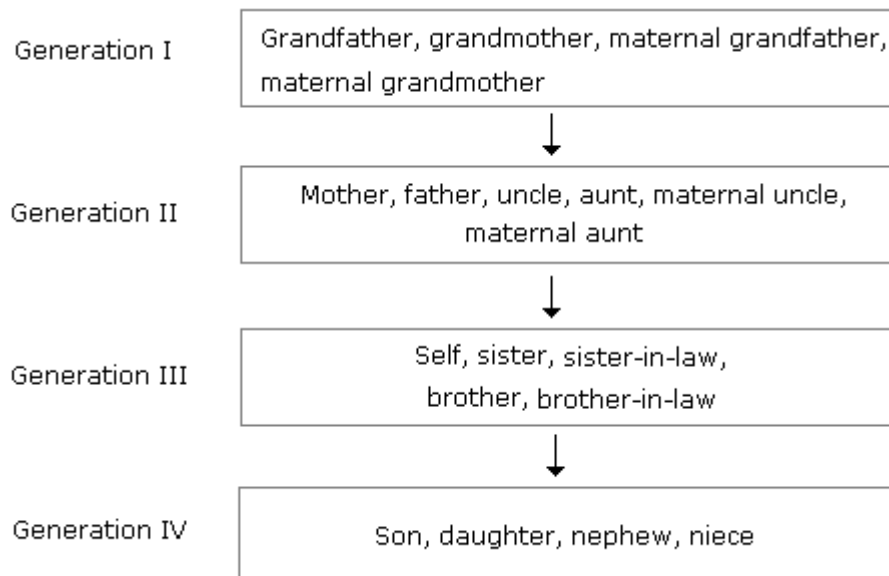
A ===== B ---> A and B are spouses

 ---> A is the Parent of B.

Conditions:

- Don't judge the genders on the name basis.
- Using the hints like "he", "she", the gender must be considered.
- Be aware of the word "Only".

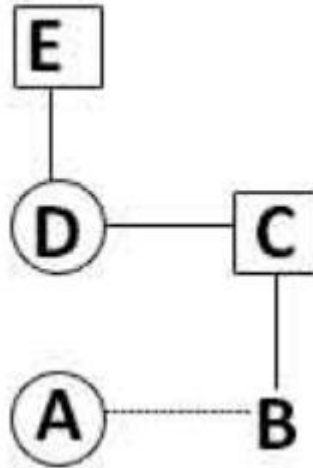
Relations from one generation to next:



Some Examples:

1) A is sister of B. C is the father of B. D is the wife of C and E is the father of D. How is E related to B?

- a) uncle b) grandmother c) Father d) Grandfather



Answer : D

Solution : From the diagram above, it is clear that,
E is grandfather of B. Hence, option D is correct answer.

2. Pointing to a man, a woman said, "He is the only son of my mother's mother". How is the woman related to the man?

- a) Aunt b) daughter c) Niece d) Sister

Answer : C

Solution: Son of my mother's mother means he is my mother's brother so that man is paternal uncle of women, and she is niece of that man. Hence, option C is correct answer.

3. CODING DECODING

Coding involves representing English alphabets with other symbols so as to hide the correct meaning. Coding has two main type of questions. i. e.

Letter Coding and Word Coding

Letter Coding: In Letter coding, an English word and its corresponding code are given. While there is a one-to-one relationship between the given letters, the kind of relationship is certainly not fixed.

- Letter Movement
- Direct Coding
- Reverse Coding

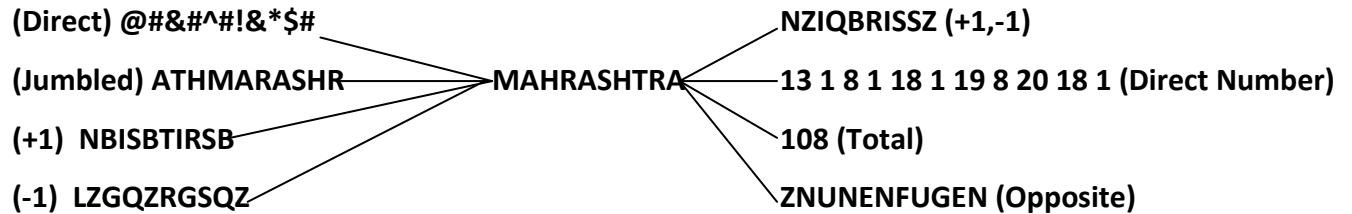
Alphabet Numbers

A	B	C	D	E	F	G	H	I	J
1	2	3	4	5	6	7	8	9	10
K	L	M	N	O	P	Q	R	S	T
11	12	13	14	15	16	17	18	19	20
U	V	W	X	Y	Z				
21	22	23	24	25	26				

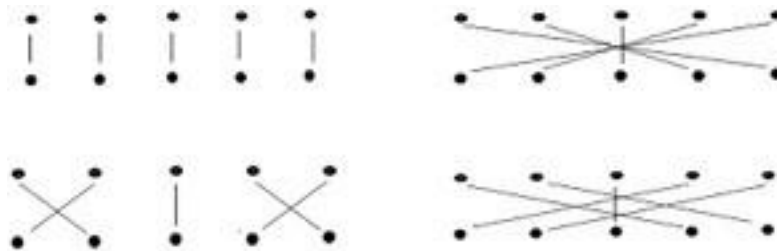
Opposite Letters

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Sample Coding



Sample Pattern



Examples

- In certain language “SIMPLE” is coded as “VLPSOH”, how is “DIFFICULT” coded in that language?

a) GLILFYOW b) G L I I L F Y O S c) GLNKPUGK d) UGKPUUGL

Answer: a) GLILFYOW

Explanation:

Add 2 letters (D+2 =G, I+2=L, F+2=I, C+2=F etc...)

- If white is called blue, blue is called red, red is called yellow, yellow is called green, green is called black, black is called violet and violet is called orange, what would be the colour of clear sky?

a)Yellow b)Red c)Blue d)None of these

Answer : Red

Explanation:

Originally color of clear sky is blue. Here Blue is coded as Red.

4. ODD ONE OUT

Odd One Out in Numbers:

Try to check for the difference between numbers or multiplication, division pattern.

Q) 36, 54, 18, 27, 9, 18.5, 4.5

a) 4.5 b) 18.5 c) 54 d) 18

Answer: b) 18.5

Solution: Alternate numbers are divided by 2

Odd One Out in Alphabets:

Q) GC, JN, MI, PT, SO

a) PT b) GC c) JN d) So

Answer: PT

Odd One Out in Words:

Q) Mars, Earth, Saturn, Sun, Neptune, Mercury

Odd One: Sun

5. SYLLOGISMS

Types of Statements

The four basic statements in syllogism are,

- All As are B (Eg. All Mobiles are cameras)
- Some As are B (Eg. Some cats are dogs)
- No A is B (Eg. No rabbit is a pig)
- Some As are not B (Eg. Some cats are not humans)

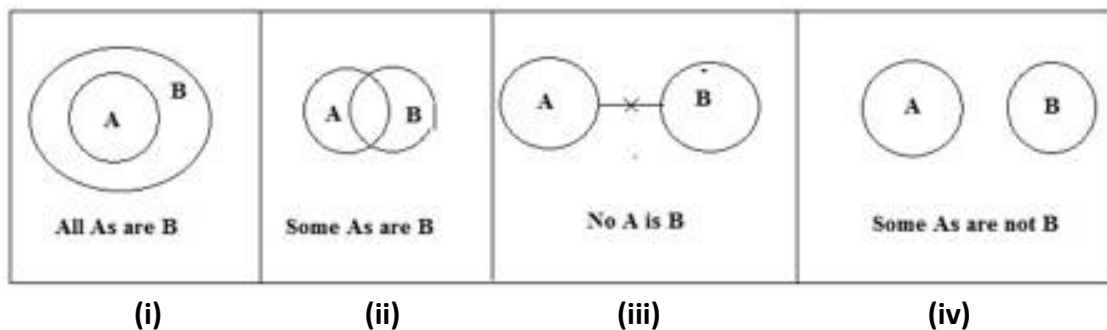
These statements can be classified into two categories as shown in below table.

	Universal	Particular
Positive	All cats are dogs	Some dogs are birds
Negative	No bird is a pig	Some pigs are not birds

Alternate Words

All, every, any, none, not a single, only etc	Universal (Positive or Negative)
Some, many, a few, quite a few, not many, very little, most of, almost, generally, often, frequently, etc.	Particular (Positive or Negative)

Basic Diagrams



For (i) statement, i.e. All As are B,

Circle A should be inside B or A and B can be equal. But circle A should not exceed B.

For (ii) statement, i.e. Some As are B, Circle A and B should be connected always. It should not separate.

For (iii) statement, i.e. No A is B, We should not connect circle A and circle B.

For (iv) statement, i.e. Some As are not B, We can connect circle A and circle B.

Complementary pair

In the Complementary pair, subject and predicate should be same in both the conclusions. If one conclusion is true, definitely the other conclusion will be false and vice versa. There are two complementary pairs in syllogism.

Pair I : All As are B & Some As are not B

If “All As are B” is true, definitely “Some As are not B” is false. If “Some As are not B” is true, definitely “All As are B” is false.

Pair II: No A is B & Some As are B

If “No A is B” is true, definitely “Some As are B” is false. If “Some As are B” is true, definitely “No A is B” is false.

Procedure:

Step 1: Draw the basic diagram for the given statements.

Step 2a: If all are positive conclusions, Check those conclusions in basic diagram and decide which one is true or false. Don't draw any other diagram if all are positive in conclusion.

Step 2b: If there is negative conclusion and it is true in basic diagram, try to make it false by drawing its complementary pair. While drawing alternate diagram, it should not violate any other given statements. If you are able to draw alternate diagram, without violating any statement. Then the negative statement is false.

Examples: (Positive Conclusions)

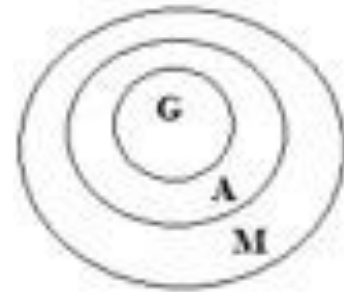
Q.1.

Statements

1. All grapes are apples
2. All apples are mangoes

Conclusions

1. All grapes are mangoes
2. All mangoes are grapes
3. Some grapes are mangoes



Answer: (1) and (3) are true

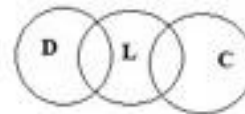
Q.2

Statements

1. Some doctors are lawyers
2. Some lawyers are circle

Conclusion

1. Some doctors are circle
2. All doctors are circle



Answer : Both (1) and (2) are false

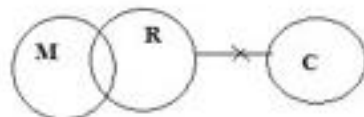
Q.3.

Statements

1. Some mobiles are rows
2. No row is circular

Conclusion

1. All circular are mobiles
2. Some circular are mobiles



Answer: Both (1) and (2) are false Explanation:

Basic diagrams for above questions. Verify the conclusion only in below basic diagram.(Because all are positive conclusion)

Examples (Negative Conclusion)

Q.1.

Statements

1. All months are weeks
2. Some week are days

Conclusions

1. No month is day
2. Some weeks are months

Answer : (2) is true

Q.2

Statements

1. All right are left
2. No left is top

Conclusion

1. Some tops are right
2. No top is right

Answer : Only (2) is true

Q.3.

Statements

1. All goats are good
2. Some good are watch

Conclusion

1. Some watch are goat
2. No watch is goat

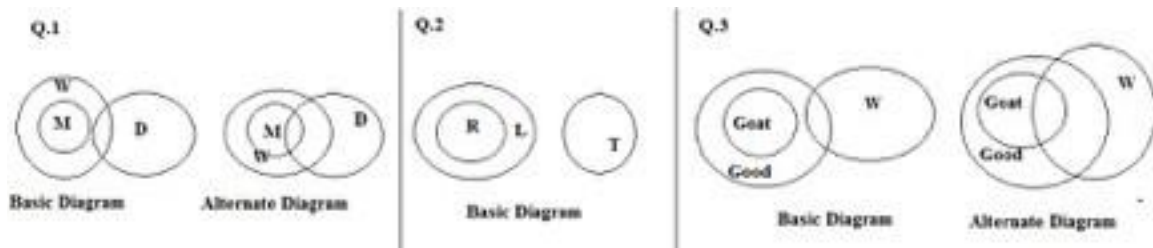
Answer : Either (1) and (2) are true

Explanations

Q.1 “No month is a day” is negative conclusion and it is true in basic diagram. So you have to make it false by drawing its complementary pair. Refer Alternate diagram. While drawing alternate diagram, it is not violating any statements. So the negative conclusion “No month is a day” is false in alternate diagram. Hence the conclusion is false.

Q.2 “No top is a right” is negative conclusion and it is true in basic diagram. So you have to make it false by drawing its complementary pair. While drawing alternate diagram, it is violating the given statements. So you can't draw alternate diagram. So the negative conclusion “No top is a right” will always true.

Q.3 “No watch is a goat” is negative conclusion and it is true in basic diagram. So you have to make it false by drawing its complementary pair. Refer Alternate diagram. While drawing alternate diagram, it is not violating any statements. So the negative conclusion “No watch is a goat” is false in alternate diagram and its complementary is true. So you have to mark either (1) or (2).



6. SEATING ARRANGEMENT

General Guidelines

1. Understand the entire question and statements quickly and correctly
2. Determine the usefulness of each information and classify them accordingly into
 - Definite Information
 - Comparative information
 - Negative Information
3. Identify the statements that give definite information.

For instance let us take three statements and evaluate them

-Statement (a): Ajay is to the left of Bhanu.

The data in the statement is basic but not definite as the statement only says that Ajay is to the left of Bhanu. But, it does not specify where Ajay is located from Bhanu.

-Statement (b): Ajay is second to the left of Bhanu.

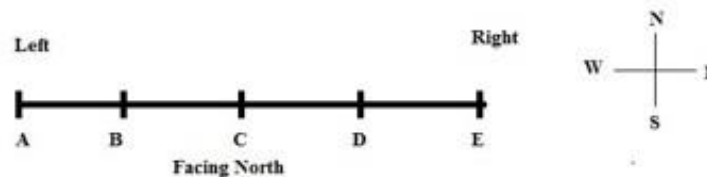
The data in the statement is definite as it clearly states that Ajay is placed second to the left of Bhanu.

-Statement (c): Tanmay is between Ramesh who plays football and Pankaj in order of seating in a row.

It can be understood as 'Tanmay is between Ramesh and Pankaj. So, they may be seated as RTP or PTR (so, the data is not definite) and 'Ramesh plays football'.

1. Search for the connecting information.
2. Figure out the seating arrangement by clearly identifying the directions

Linear Arrangement



If they are facing North,

1. B,C,D,E are right of A but only B is immediate right of A.
2. D,C,B,A are left of E but only D is the immediate left of E.
3. A is the immediate left of B while E is the immediate right of D.

If they are facing South,

1. B,C,D,E are left of A but only B is immediate left of A.
2. D,C,B,A are right of E but only D is the immediate right of E.
3. A is the immediate right of B while E is the immediate left of D.

Arrangement Puzzle

Tips for how to solve Arrangement Puzzle

Focus on Family-tree first: You can't successfully start placing people in the seating arrangement if you don't know the family tree. Usually people in the seating arrangement are referred to as: wife of C, son of D. So it is better if you know who is who and then start with the seating arrangement. To do this, read all conditions in the question carefully from the beginning and ignore all statement about who is sitting where.

Find starting points: Starting Points help in identifying the position of specific persons. Generally, the question begins with negative information which doesn't highlight the exact position. Remember, statements starting with "Neither/Nor" can only reveal relative positions and can never be a starting point. You must read all the statements carefully to be able to arrive at the starting point.

Find connecting dots: Connecting dots are small pieces of information which help in the further arrangement based on the position of people already seated.

Solve questions purely on blood relations first: If you are unable to solve the final seating arrangement but have already drawn the family tree, the answer the relevant questions. Don't leave the entire block just because you couldn't solve it completely.

ARRANGEMENTS

Seating arrangement is arranging people in their perspective position based on the data specified in the given question. Arrangements can be made in any form like row, circle, triangle, square, rectangle etc. The position can be inferred by the direction and that directions determine the types of seating arrangement

1. Linear Arrangement
2. Circular Arrangement
3. Rectangular / Square Arrangement
4. Hexagonal Arrangement

In order to solve seating arrangement questions, first of all diagram should be made. By doing so questions are easily and quickly solved.

7.DATA SUFFICIENCY

Data Sufficiency problem consists of a question followed by two statements, labeled as Statement (1) and Statement (2), in which certain data are given. You have to decide whether the data given in the statements are sufficient for answering the question.

Using the data given in the statements upheld your knowledge of mathematics and everyday facts (such as the number of days in July or the meaning of counterclockwise), you must indicate from the following answer options that whether:

1. Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient to answer the question asked;
2. Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient to answer the question asked;
3. Both statements (1) and (2) TOGETHER are sufficient to answer the question asked; but NEITHER statement ALONE is sufficient.
4. EACH statement ALONE is sufficient to answer the question asked;
5. Statements (1) and (2) TOGETHER are NOT sufficient to answer the question asked, and additional data specific to the problem are needed.

A statement is sufficient when it guarantees exactly one answer to that question.

TIPS TO SOLVE DATA SUFFICIENCY PROBLEMS:

- Never try to reach final answer as it is not asked. You need to find whether the information provided is enough to solve the given problem or not.
- Never make any assumption. Use only universal rules { eg. $a + b = a + b - (a \cup b)$ }
- Try to solve questions by using above strategies
- Solve question step by step. First try to find answer using first statement then second and finally with both then mark the answer
- Even if you find answer with only one statement, then try to find answer with remaining statement as sometimes there is an option that answer can be find with both statements separately.
- Move on quickly and mark answer can't be found in case you are unable to reach any conclusion with information provided.

The Data Sufficiency questions do not require the exam aspirant to find the exact answer. Infact the only thing that they require is to determine whether the statements provided in the question contain enough information for answering it.

Example 1: A, B, C, D, and E scored different marks in different exams. From statements given below determine who has secured the last position.

Statement I: D has scored fewer marks than C and E.

Statement II: B has scored more marks than A.

Options:

1. Data in statement 1 is sufficient alone to determine the answer.
2. Data in statement 2 is sufficient alone to determine the answer.
3. Data in either of the statements is sufficient to determine the answer.
4. Data provided in both the statements together are not sufficient to determine the answer.
5. Data in both statements are required to determine the answer.

Solution:

From the statement I, you can determine that marks scored by D < marks scored by C and E. While from the statement II you can see that marks scored by N > marks scored by M. No statement alone can determine who has scored the least marks. Using two statements together you can see that either D or M must have scored the least marks. But there is no definite answer. And thus option (4) is the correct answer.

Example 2: What is Reena's rank in the class?

I. There are 26 students in the class.

II. There are 9 students who have scored less than Reena.

- (1) if the data in statement I alone are sufficient to answer the question;
- (2) if the data in statement II alone are sufficient to answer the question;
- (3) if the data either in I or II alone are sufficient to answer the question;
- (4) if the data even in both the statements together are not sufficient to answer the question;
- (5) if the data in both the statements together are needed;

Exp: Option (5)

From I and II, we conclude that there are 16 students above Reena in rank. Thus, Reena's rank is 17th in the class. So, both the statements are necessary.

8.MATHEMATICAL INEQUALITY

Mathematical Inequality (Positive)

Symbol	Meaning	Example
$>$	Greater than	$A > B$ -- A is greater than B
$<$	Less than	$A < B$ -- A is less than B
$\leq$	Less than or equal to	$A \leq B$ -- A is less than or equal to B
$\geq$	Greater than or equal to	$A \geq B$ -- A is less than or equal to B
$=$	Equal to	$A = B$ -- A is equal to B

Mathematical Inequality (Negative)

Symbol	Meaning	Example
$\nmid >$	Not Greater than	$\leq$
$\nmid <$	Less than	$\geq$
$\nmid \leq$	Not less than or equal to	$<$
$\nmid \geq$	Not greater than or equal to	$>$
$\neq$	Not equal to	$< \text{ or } >$

Tips to solve inequality

1. $A > B \geq C \implies A > C$

2. $A \geq B > C \implies A > C$

3. $A > B = C \implies A > C$

4. $A = B > C \implies A > C$

5. $A < B \leq C = D \implies A < D \text{ and } B \leq D$

6. $A < B \leq C > D = E \implies A < C \text{ and } C > E$

In this case, the relations between AD, AE, BD and BE cannot be established. For e.g. $A < C$ and $C > D$ so we get $A < C > D$. That means C is greater than both A and D. But we don't know which is greater—A or D; or if they are both equal. Thus the relation between A and D cannot be established.

7. $A > B \leq C ! \geq D \leq E \implies A > B \leq C < D \leq E \implies B < E, C < E, B < D.$

But the relations between AC, AD, and AE cannot be established.

9. STATEMENT AND COURSES OF ACTION

The questions from topic '**Statement & Course of action**' are asked to test the ability of aspirant to **judge a problem correctly** as well as to determine the **root cause of the problem** and then to **prescribe a suitable course of action**.

What is the Course of Action?

A course of action is a step or **administrative decision** to be taken up for improvement or follow-up for further action in regard to the **problem, policy** etc. on the basis of the information given in the statement.

Broadly, we can classify these questions in the following two types:

1. Problem and Solution based: When the given situation talks of a problem, then the suggested 'course of action' should talk about the solution keeping in mind the following points-

- It solves/reduces or minimizes the problem.
- The solution or course of action is practically Possible.

Note:

1. A suggested course of action may indeed solve a problem but in **practical life, it may not be advisable or possible**. If it is so, **then the course of action is rejected**.

2. When the statement is an established fact. ie, It is acknowledged universally as a fact.

For e.g. –

Statement: A large number of people in ward X of the city are diagnosed to be suffering from a fatal malaria type.

Courses of Action: I. The city municipal authority should take immediate steps to carry out extensive fumigation in ward X.

1. The people in the area should be advised to take steps to avoid mosquito bites.

(1). Only I follows

(2). Only II follows

(3). Either I or II follows

(4). Neither I nor II follows

(5). Both I and II follow

The answer is (5) – Clearly, prevention from mosquitoes and elimination of mosquitoes are two ways to prevent malaria. The action will reduce the problem. So, both the courses of action follow.

2. Fact and Improvement based: When the presented situations talk about a simple fact (**not a problem, just a situation**) then the suggested courses of action should suggest ways of improvement.

The format of the Problem: These types of questions contain two or more statements and these statements are followed by a number of Courses of Action. You are required to find which Course of Action will logically follow from the given statement.

For e.g.-

Statement: Exporters in the capital are alleging that commercial banks are violating a Reserve Bank of India directive to operate a post-shipment export credit denominated in foreign currency at international rates from January this year.

Courses of Action: I. The officers concerned in the commercial banks are to be suspended.

II. The RBI should be asked to stop giving such directives to commercial banks.

(1) Only I follows

(2) Only II follows

(3) Either I or II follows

(4) Neither I nor II follows

(5) Both I and II follow

Answers (4) – The statement mentions that the commercial banks violate a directive issued by the RBI. The remedy is only to make the banks implement the Act. So, none of the courses follows.

Steps to Solve Statement and Course of Action Questions

Following are the approaches to solve the questions of this topic :

1. Don't make your individual perception.
2. Act according to the rules of any organization.
3. Go for impartial and not personal thinking.
4. Existing practices are not effective Course of Action.
5. Whenever a problem arises, it is always a sensible course of action to find out the cause.

Important Points to proceed step by step to reach final courses of action:

1. Extreme or strict action is not a valid course of action.
2. The actions should be a positive step towards the solution of the problem, rather harsh and undemocratic.
3. Analyses course of action whether it will solve the problem or lessen it. For proper analysis check with

- (i) Universal truth
- (ii) Experience
- (iii) Logic

4. Go for a practical course of action. A simple problem must have a simple course of action, not a complex one which might create more problem than to solve or reduce.

5. In most of the cases, a situation has more than one course of action. But they are never exclusive to each other. So our answer should be always “both follow” and not “either of them follows”.

6. If the course of action and problems are properly balanced and if the course of action is also feasible than that action can be followed.

While solving the problems on course of action, the following points should be considered.

#Have a neutral and impartial judgment

#Identify the main problem (root cause) and apply logic to figure out the problem-solving steps (courses of action)

#Course of action can't be too extreme

#Identify the established fact.

#Identify if the steps to be taken (course of action) are logically possible.

#Identify the experienced truth.

REMEMBER:

A course of action must be able to

- *Solve the problem*
- *Reduce the problem*
- *Minimize the problem*
- *Subside the problem*

10. STATEMENT AND ASSUMPTIONS

The questions from statements and assumptions feature regularly in competitive exams. In these questions, a statement is given in the question and it is followed by two or more assumptions. For the correct answer, the student needs to assess the given statement and decide which of the given assumption is implicit to the statement in the question above. To solve the questions you need to follow a pattern and thus it becomes essential to know what statements and assumptions stand for.

The following points should be considered while solving questions based on statement and conclusion:

1. Assumptions should be drawn only from the given statement, not from your real life experiences
2. Don't extrapolate beyond what is given in the statement to arrive at false assumptions
3. If a notice/ advertisement is put up, the implicit assumption is that it will have impact

Let us understand this concept by considering an example:

Q.1.

Statement: "In order to bring punctuality in our office, we must provide conveyance allowance to our employees." – In charge of a company tells Personnel Manager.

Assumptions:

Conveyance allowance will not help in bringing punctuality.

Discipline and reward should always go hand in hand.

- A. Only assumption I is implicit
- B. Only assumption II is implicit
- C. Either I or II is implicit
- D. Neither I nor II is implicit
- E. Both I and II are implicit

Answer: Option B

Explanation:

Assumption I goes against the statement. So, it is not implicit. The allowance will serve as a reward to the employees and shall provoke them to come on time. So, II is implicit.

11.Logical Series

Series is a sequential order of letters, numbers or both arranged in such a way that each term in the series is obtained according to some specific rules. These rules can be based on mathematical operations, place of letters in alphabetical order etc.

Different types of questions covered in this chapter as follows

- **Number Series**
- **Letter Series**
- **Alpha-Numeric Series**
- **Continuous Pattern Series**

The candidates are required to identify the pattern involved in the formation of series and accordingly find the missing term to complete the series. Also, there may be some questions where one of the terms in the series is incorrect and the candidate is required to find out that term of the series by identifying the pattern involved in the formation of series.

There is no set pattern and each question may follow a different pattern or sequential arrangement of letters or digits, which you have to detect using your common sense and reasoning ability.

Let us understand this concept by considering some Examples:

Que 1. Look at this series: 8, 6, 9, 23, 87, ... What number should come next?

- A. 128
- B. 226
- C. 324
- D. 429

Answer: Option D

Explanation:

$$8 \times 1 - 2 = 6$$

$$6 \times 2 - 3 = 9$$

$$9 \times 3 - 4 = 23$$

$$23 \times 4 - 5 = 87$$

$$87 \times 5 - 6 = 429....$$

Que.2) JAK, KBL, LCM, MDN, _____

- A. OEP
- B. NEO
- C. MEN
- D. PFQ

Answer: Option B

Explanation:

This is an alternating series in alphabetical order. The middle letters follow the order ABCDE. The first and third letters are alphabetical beginning with J. The third letter is repeated as a first letter in each subsequent three-letter segment.

Que.3) Study the following digit-letter-symbol sequence carefully and answer the questions given below:

4 P + S r 9 B # A 3 ? 7 c > Z % 6 d & Q @ 1

If the above sequence is written in the reverse order, then which element will be 5th to the right of 16th element from the right end?

Solution: Reversed Sequence - 1 @ Q & d 6 % Z > c 7 ? 3 A # B 9 r S + P 4.

Hence, now the 16th element from the right end is % and 5th to the right of % is ?

Shortcut method: There is an alternate way to find the answer without writing the sequence in reverse order. Since both the directions are given from right, subtract them: $16R - 5R = 11R$

Now, as the questions asks the sequence to be reversed, the element is 11th from left of the original series, i.e. '?'.

Que.4)_bcc _ ac _ aabb _ ab _ cc

- A. aabca
- B. abaca
- C. bacab
- D. bcaca

Answer: Option C

Explanation:

The series is bbccaa/ccaabb/aabbcc. Thus, the letter pairs move in a cyclic order.

12.Clocks

1. A clock has two hands, the smaller one is called the hour hand and the larger one is called minute hand.
2. Angle traced by minute hand in 1 hr = 360° .
3. Angle traced by hour hand in 1hr = 30°
4. Speed of minute hand = 6° per min
5. Speed of hour hand = $(\frac{1}{2})^\circ$ per min
6. The hour hand and minute hand coincide 22 times in a day. (Refer below picture)
7. The hour hand and minute hand will be in straight line and opposite (180°) 22 times in a day.
8. The hour hand and minute hand will be in right angle (90°) 44 times in a day.

FORMULAS

To calculate angle between hour hand and minute hand

$$\text{Angle} = \left(\frac{11}{2}M - 30H\right)$$

To calculate time during coincide, straight line and right angle.

- | | | |
|-----------------------------|---|--|
| 1. Straight line (Coincide) | : | (5H+0) 12/11 (or) 60H/11 |
| 2. Right angle | : | (5H+15) 12/11 minutes past H |
| 3. Straight line (Opposite) | : | a. (5H-30) 12/11 minutes past H, when $H > 6$
b. (5H+30) 12/11 minutes past H, when $H < 6$ |

TYPES OF PROBLEMS

1. Calculating the angle between Hour Hand and Minute Hand
2. Calculating the exact time during Straight line, Coincide & Right angle
3. Problems on incorrect clocks

1. What will be the angle between hour hand and minute hand at 7.50?

- a) 45° b) 56° c) 65° d) 75°

Answer: c) 65°

Solution : $(11 \times 50/2) - (30 \times 7)$

$$= 275 - 210 = 65^\circ$$

2. At what time between 7 and 8'o clock will the hands of a clock in the same straight line but, not together?

a) 5 min. past 7 b) $5\frac{2}{11}$ min. past 7 c) $5\frac{3}{11}$ min. past 7 d) $5\frac{5}{11}$ min. past 7

Answer: d) $5\frac{5}{11}$ min. past 7

Solution: $(5 \times 7 - 30) \times \frac{12}{11} = 60/11 = 5\frac{5}{11}$

3. A watch gains 5 seconds in 3 minutes and was set right at 8 AM. What time will it show at 10 PM on the same day?

a) 10:20:20 PM b) 10:23:20 PM c) 10:20:23 PM d) 11 : 30 PM

Answer: b) 10 : 23 : 20 PM

Solution:

The watch gains 5 seconds in 3 minutes So 100 seconds in 1 hour. From 8 AM to 10 PM on the same day, time passed is 14 hours. In 14 hours, the watch would have gained 1400 seconds or 23 minutes 20 seconds. So, when the correct time is 10 PM, the incorrect watch would show 10 : 23 : 20 PM